Week 4 — Session 1

Topic: Constraint-Driven Solution Design

Language: C++17

Focus: using constraints and frequency counting to choose an efficient approach.

Task 1: Determine if String Halves Are Alike

Practice: https://leetcode.com/problems/determine-if-string-halves-are-alike/

Split the string into two equal halves and compare the number of vowels in each half.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

bool isVowel(char c) {
    c = tolower(static_cast<unsigned char>(c));
    return c == 'a' || c == 'e' || c == 'i' ||
           c == 'o' || c == 'u';
}

bool halvesAreAlike(const string& s) {
    int n = s.size();
    int left = 0, right = 0;

    for (int i = 0; i < n / 2; ++i) {
        if (isVowel(s[i])) left++;
        if (isVowel(s[i + n / 2])) right++;
    }

    return left == right;
}

int main() {
    string s;
    cin >> s;

    cout << (halvesAreAlike(s) ? "true" : "false") << '\n';
    return 0;
}
```

Task 2: LAPIN — Lapindromes

Practice: https://www.codechef.com/problems/LAPIN

For an even-length string, compare the character frequencies in the left and right halves. For an odd-length string, ignore the middle character.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

bool isLapindrome(const string& s) {
    int n = s.size();
    vector<int> left(26, 0), right(26, 0);

    for (int i = 0; i < n / 2; ++i)
        left[s[i] - 'a']++;

    for (int i = (n + 1) / 2; i < n; ++i)
        right[s[i] - 'a']++;

    return left == right;
}

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int T;
    cin >> T;

    while (T--) {
        string s;
        cin >> s;
        cout << (isLapindrome(s) ? "YES" : "NO") << '\n';
```

```
    }
    return 0;
}
```

All programs are standalone C++17 solutions and can be compiled with a standard C++17 compiler.